



## University of Central Punjab

### Faculty of Information Technology

**PROGRAM (S) TO BE  
EVALUATED**

**BSCS**

#### **A. Course Description**

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Course Code</b>                                            | AL352                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Title</b>                                           | Artificial Intelligence – Lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Credit Hours</b>                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Prerequisites by Course(s) and Topics</b>                  | Data Structures and Algorithms<br>Discrete Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Assessment Instruments with Weights</b>                    | Graded Labs (20)<br>Mid Term (20)<br>Final Term (45)<br>Project (10)<br>Class Participation (5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Instructor</b>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Coordinator</b>                                     | Dr. Mohsin Ashraf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Office Hours</b>                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Plagiarism Policy</b>                                      | All the parties involved in first cheating case will be awarded Zero for that evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Current Catalog Description</b>                            | This course will introduce the basics of artificial intelligence (AI), its scope and application domain. Topics such as knowledge representation, strategies for the State space search, Heuristic Search, first order logic, knowledge-based systems and state of the art computational techniques like neural networks, genetic algorithm will be covered during the course.                                                                                                                                                                                                                                       |
| <b>Textbook (or Laboratory Manual for Laboratory Courses)</b> | Artificial Intelligence: A modern Approach by Russell and Norvig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Reference Material</b>                                     | Artificial Intelligence Illuminated by Ben Coppin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Course Goals</b>                                           | AI is a field that studies how to realize the intelligent human behaviors on a computer. The ultimate goal of AI is to make a computer that can learn, plan, and solve problems autonomously. Understand the various searching techniques, example problems-game playing techniques.<br>Apply these techniques in applications which involve perception, reasoning and learning. Explain the role of agents and how it is related to environment and the way of evaluating it and how agents can act by establishing goals. Acquire the knowledge of real-world Knowledge representation. Analyze and design a real- |

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|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------|------------------------|----------------------------------|
|                                                                            | world problem for implementation and understand the dynamic behavior of a system. |                         |                        |                                  |
| <b>Topics Covered in the Course, with Number of Lectures on Each Topic</b> | Attached                                                                          |                         |                        |                                  |
| (assume 15-week instruction and one-hour lectures)                         |                                                                                   |                         |                        |                                  |
| <b>Programming Assignments Done in the Course</b>                          | Yes                                                                               |                         |                        |                                  |
| <b>Lab Time Spent on (in credit hours)</b>                                 | <b>Theory</b>                                                                     | <b>Problem Analysis</b> | <b>Solution Design</b> | <b>Social and Ethical Issues</b> |
|                                                                            | 0.5                                                                               | 1                       | 1.5                    |                                  |
| <b>Oral and Written Communications</b>                                     | Yes                                                                               |                         |                        |                                  |

| <b>CLO</b> | <b>CLO STATEMENT</b>                                                                                                         | <b>Bloom's Taxonomy Level</b> | <b>PLO</b> |
|------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------|
| 1          | Design and implement AI-based solutions to computational problems using appropriate techniques and tools in lab experiments. | C6                            | 4          |

| Week | Lecture | Topics Covered                                                                                                                                                                                                                                                                                                            |
|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 1       | <b>Introduction to Artificial Intelligence:</b> History, Goals, Applications, Different AI Approaches, Turing Test, Cognitive models, Syllogism, Rational Agents                                                                                                                                                          |
|      | 2       | <b>Intelligent Agents,</b> The Concept of Rationality, The Nature of Environments, The Structure of Agents                                                                                                                                                                                                                |
| 2    | 3       | <b>Solving Problems by Searching,</b> Problem-Solving Agents, Problem Formulation, Search spaces, State space, Search Trees, Initial and Goal State, Successor Function, Objective Function, Goal Test, Goal Path, Example Problems (Missionaries & Cannibals, Wolf Duck and Farmer, 8-Puzzle, Tic-Tac-Toe, 8-Queen, TSP) |
|      | 4       | <b>Uninformed/Blind Search:</b> Breadth First Search (BFS), Depth First Search (DFS), Time Complexity, Space Complexity, Completeness, Optimality                                                                                                                                                                         |
| 3    | 5       | <b>Uninformed/Blind Search:</b> Depth Limited Search, Uniform Cost Search, IDS, Algorithm Complexity                                                                                                                                                                                                                      |
|      | 6       | Implementation Details on Example Problems                                                                                                                                                                                                                                                                                |
| 4    | 7       | <b>Informed searches:</b> Heuristics, Heuristic Functions and Properties, Best First Search, Greedy Search                                                                                                                                                                                                                |
|      | 8       | <b>Informed Searches:</b> , A* Search,                                                                                                                                                                                                                                                                                    |
| 5    | 9       | <b>Informed Searches:</b> Monotonicity and Admissibility of Heuristics, Optimality vs Efficiency                                                                                                                                                                                                                          |
|      | 10      | <b>Informed Searches:</b> IDS A*                                                                                                                                                                                                                                                                                          |
| 6    | 11      | Implementation Details on Example Problems                                                                                                                                                                                                                                                                                |
|      | 12      | <b>Meta-Heuristics and Local Search,</b> Hill Climbing, Local minima and maxima, Variants of Hill Climbing Algorithm                                                                                                                                                                                                      |
| 7    | 13      | <b>Adversarial Search:</b> Game Trees, Minimax                                                                                                                                                                                                                                                                            |
|      | 14      | Alpha beta pruning                                                                                                                                                                                                                                                                                                        |
| 8    | 15      |                                                                                                                                                                                                                                                                                                                           |
|      | 16      | Review                                                                                                                                                                                                                                                                                                                    |
| 9    | 17      | Genetic Algorithms, Representation, Fitness, Selection                                                                                                                                                                                                                                                                    |
|      | 18      | Mutation and Cross over, Solution of 8-queens , TSP through Genetic Algorithms                                                                                                                                                                                                                                            |
| 10   | 19      | <b>Knowledge Representation (KR):</b> Logic based, procedural, structural and Network representations, Frames                                                                                                                                                                                                             |
|      | 20      | <b>Propositional Logic,</b> Syntax, Connectives, Sentences, Ambiguity in Grammar, Semantics,                                                                                                                                                                                                                              |
| 11   | 21      | <b>Models, Truth Tables, Wumpus World</b>                                                                                                                                                                                                                                                                                 |
|      | 22      | <b>Inference Rules, Practice Session for application of Inference Rules,</b>                                                                                                                                                                                                                                              |
| 12   | 23      | <b>FOL, Predicate Logic,</b> Problems with propositional logic, Predicates, Universal Quantifiers, Existential Quantifiers                                                                                                                                                                                                |
|      | 24      | <b>Man wolf and Goat etc., Skolemization, Substitution, Unification ,</b>                                                                                                                                                                                                                                                 |

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| 13 | 25 | <b><u>Introduction to Artificial Neural Networks</u></b> , Supervised and Unsupervised Learning, Perceptron                               |
|    | 26 | AND, OR classification through Perceptron                                                                                                 |
| 14 | 27 | <b><u>Supervised Learning</u></b> : Bayesian Classification, Naïve Bayes Theorem, Conditional Independence of attributes<br>Decision tree |
|    | 28 | K-NN, Working example of Naïve Bayes for classification example                                                                           |
| 15 | 29 | Clustering with K-means                                                                                                                   |
|    | 30 | Review                                                                                                                                    |